

Capture the Flag 6: Active or Inactive?

Goal: Make your Student Directory respond to student data by conditionally showing, hiding, and changing UI elements based on each student's state.

Instructor Note

These activities require your own problem-solving and research skills.

DO NOT use AI to generate the code for you.

You may use AI to:

- Research what a Dart/Flutter feature does
- Understand syntax
- Explain an error (not fix your error)
- Clarify documentation

You may **NOT** ask AI to generate the solution/code for a flag.

Flag 0 — Create Your Feature Branch

Objective: Create a new feature branch before adding conditional UI.

Create:

`feature/conditional-ui`

Screenshot:

- Terminal showing your current branch

```
satom@Tomilings MINGW64 ~/OneDrive - TAFE NSW/IT314AB_Verano (feature/student-interactions)
$ git checkout -b feature/conditional-ui
Switched to a new branch 'feature/conditional-ui'

satom@Tomilings MINGW64 ~/OneDrive - TAFE NSW/IT314AB_Verano (feature/conditional-ui)
$ git branch
* feature/conditional-ui
  feature/data-ui
  feature/exam
  feature/exam_2
  feature/student-interactions
  feature/student-list
  feature/widgets
  main
```

🚩 Flag 1 — Show Active or Inactive

Objective: Change the Student Card based on whether a student is active or inactive.

Add an **active** status to your student data.

Research how Flutter uses **conditional expressions** to display different widgets.

Example states:

- Active
- Inactive

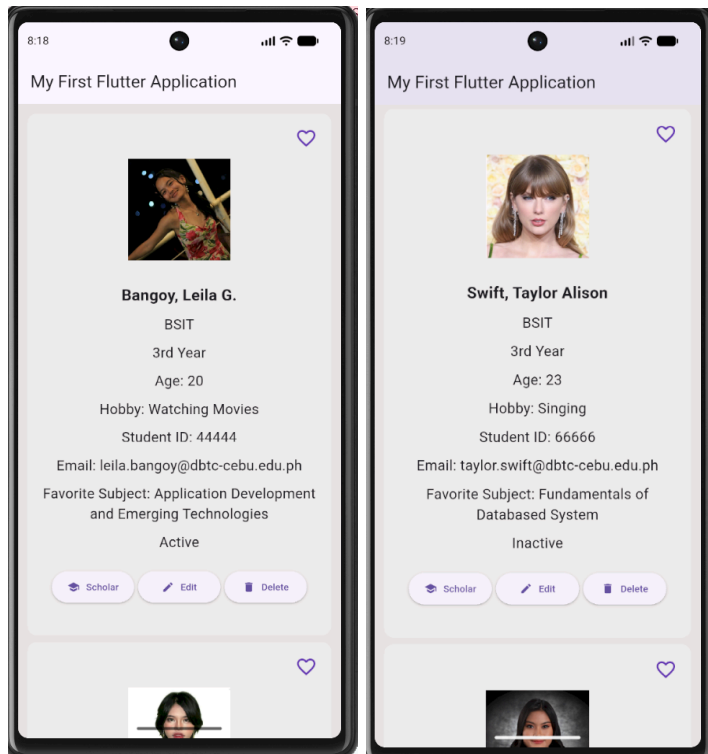
Test

Active students should display an Active indicator.

Inactive students should display an Inactive indicator.

Screenshot:

- Active Student Card
- Inactive Student Card
- Conditional logic in your code



```

198
199
200
201
202
203
204
Text(
  profile.active ? "Active" : "Inactive",
  textAlign: TextAlign.center,
  style: TextStyle(fontSize: 18),
), // Text
  SizedBox(height: 20),

```

🚩 Flag 2 — Show or Hide

Objective: Display certain UI elements only when a condition is true.

Add a conditional element to the Student Card.

Examples:

- Favorite badge appears only when favorited
- Warning appears only when inactive
- Delete action appears only if active

Research ways to conditionally include or exclude widgets from the widget tree.

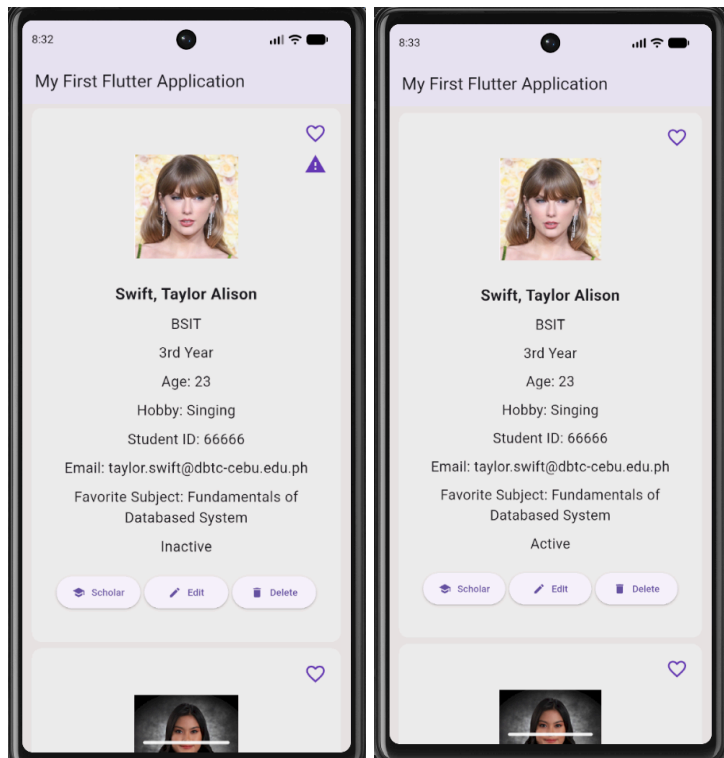
Test

When the condition is true, the element appears.

When the condition is false, the element disappears.

Screenshot:

- UI with the element visible
- UI with the element hidden
- Conditional widget code



```
332  
▲ 333      if (!profile.active)  
          Icon(Icons.warning, size: 28, color: Colors.deepPurple),
```

🚩 Flag 3 — The Empty Directory

Objective: Handle situations where there are no students to display.

Modify your Student Directory so it detects when the student list is empty.

Create an **empty state**.

Examples:

- "No students found"
- Empty-state icon
- Button to add a student

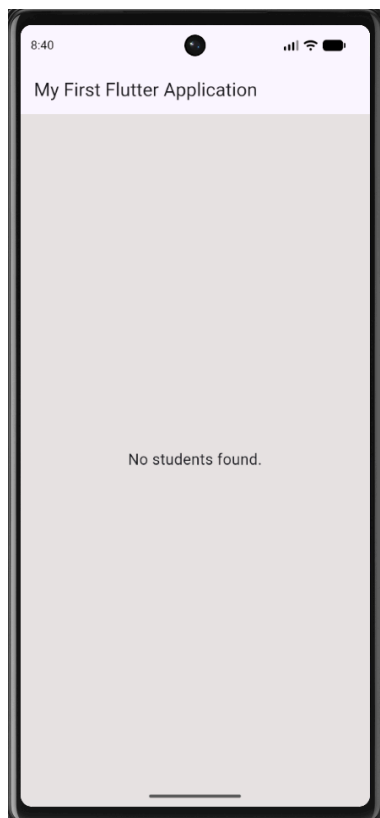
Test

Delete all students in your application (not the code)

Instead of showing a blank screen, your application should display your empty state.

Screenshot:

- Empty student list
- Empty-state UI
- Condition checking for an empty list



```
31 List<Profile> profiles = [];
```

```
351 body: profiles.isEmpty
352   ? const Center(
353     child: Text(
354       "No students found.",
355       style: TextStyle(fontSize: 18),
356     ), // Text
357   ) // Center
358 : ListView.builder(
```


Objective: Display different UI while student data is being loaded.

Create a loading state for your Student Directory.

Research how Flutter can conditionally display a loading indicator.

Examples:

- `CircularProgressIndicator`
- Loading message
- Loading placeholder

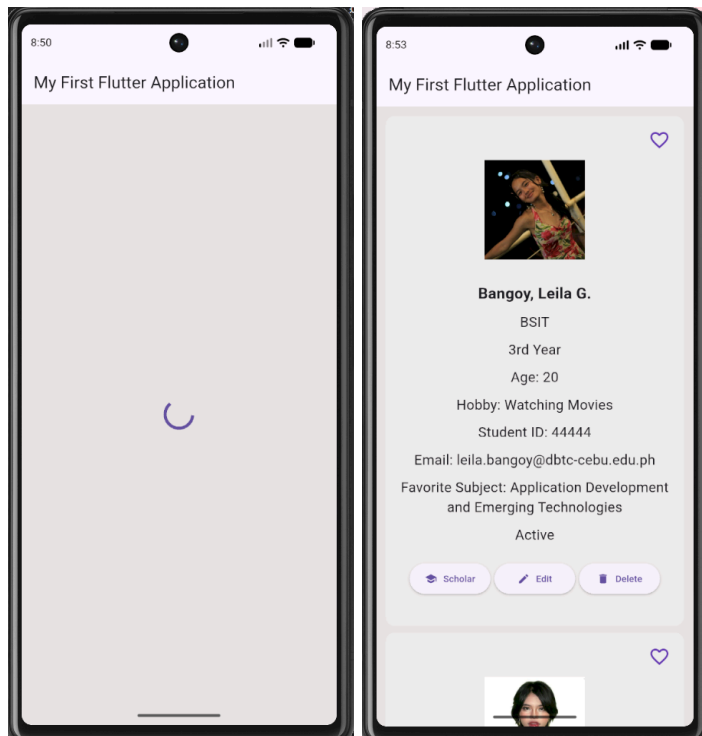
Test

When the application is in a loading state, the student list should not be displayed as the main content.

Once loading is complete, the Student Directory should appear.

Screenshot:

- Loading state
- Loaded state
- Conditional loading logic



```
353         body: isLoading
354         ? const Center(child: CircularProgressIndicator())
355         : profiles.isEmpty
356         ? const Center(
357             child: Text(
358                 "No students found.",
359                 style: TextStyle(fontSize: 18),
360             ), // Text
361         ) // Center
```

Flag 5 — Multiple States

Objective: Make your Student Directory handle multiple UI states correctly.

Your application should now be able to display:

1. Loading
2. Students available
3. No students available

Determine which state should be displayed based on your data.

Test

Verify each state:

Loading

→ Loading UI

Students available

→ Student List

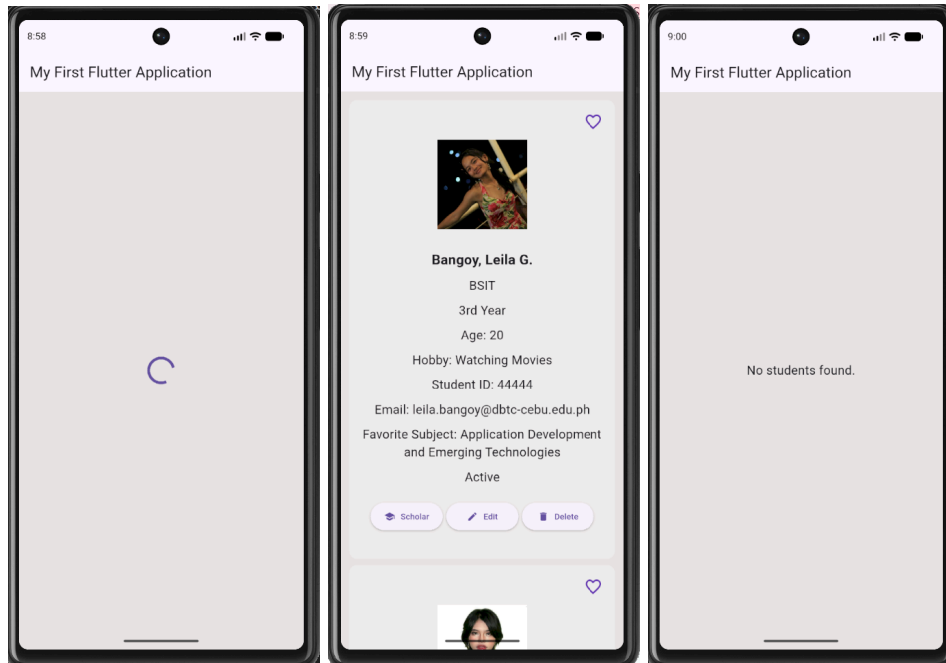
Empty list

→ Empty State

Only the appropriate UI should appear at a time.

Screenshot:

- Loading state
- Student list state
- Empty state



🚩 Flag 6 — States Work Together

Objective: Make sure conditional UI does not break your existing interactions.

Test the combination of:

- Favorite
- Delete
- Edit
- Active/Inactive
- Show/Hide
- Empty state
- Loading state

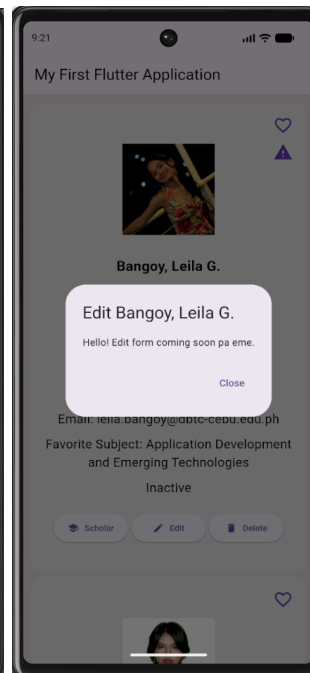
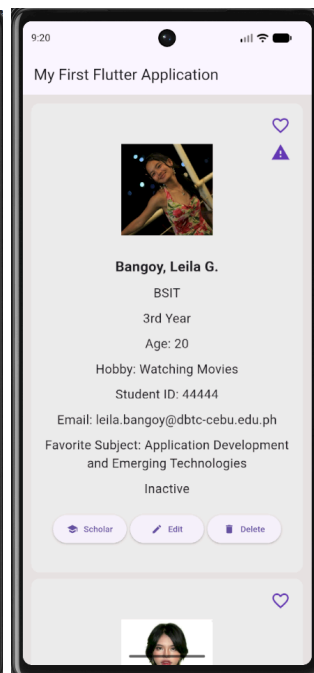
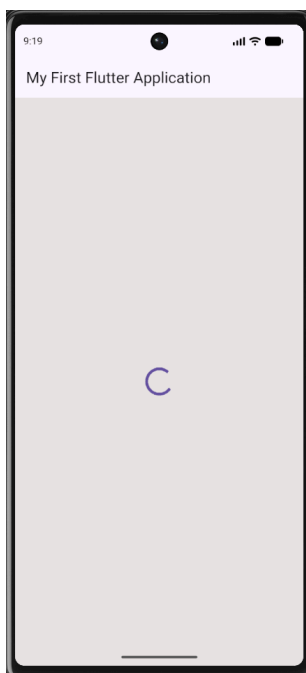
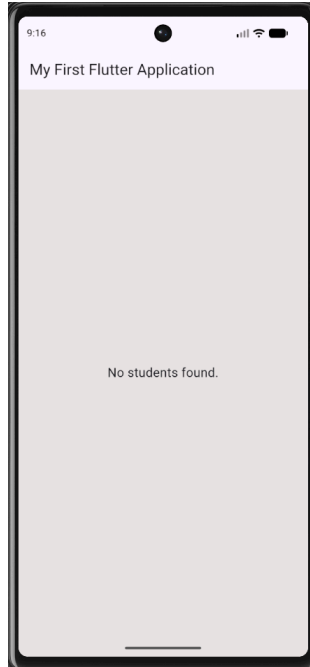
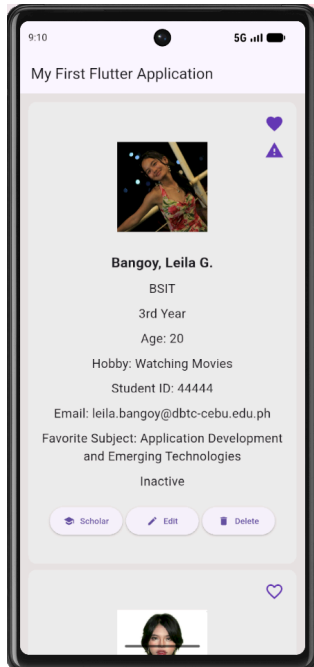
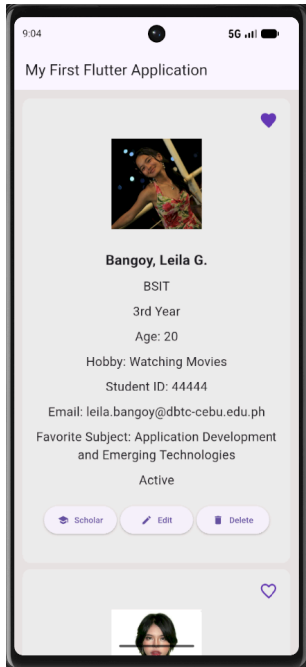
Test Scenario

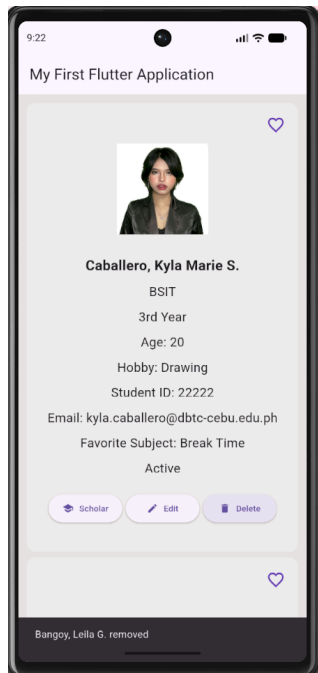
1. Favorite a student.
2. Change or display their Active/Inactive state.
3. Delete students until the directory becomes empty.
4. Verify the Empty State appears.
5. Reload or restore students.
6. Verify the Student Directory appears again.

The application should continue functioning correctly as its state changes.

Screenshot:

- Multiple conditional states working
- Interactive Student Card
- Empty State
- Application functioning normally





🚩 Flag 7 — Save Your Progress

Objective: Commit and push your completed Conditional UI Student Directory.

Use a proper Git commit message.

Branch:

`feature/conditional-ui`

Screenshot:

- Successful commit
- Current branch
- Terminal showing the commit

MINGW64~/Users/satom/OneDrive - TAFE NSW/IT314AB_Verano

satom@Tomilings MINGW64 ~/OneDrive - TAFE NSW/IT314AB_Verano (feature/conditional-ui)

```
1 ui)
$ git branch
* feature/conditional-ui
  feature/data-ui
  feature/exam
  feature/exam_2
  feature/student-interactions
  feature/student-list
  feature/widgets
  main
```

satom@Tomilings MINGW64 ~/OneDrive - TAFE NSW/IT314AB_Verano (feature/conditional-ui)

```
$ git status
On branch feature/conditional-ui
Your branch is up to date with 'origin/feature/conditional-ui'.
```

```
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <files>..." to discard changes in working directory)
        modified:   FlutterProjects/my_first_flutter_app/lib/main.dart
        modified:   FlutterProjects/my_first_flutter_app/linux/Flutter/generated_plugin_registrant.cc
        modified:   FlutterProjects/my_first_flutter_app/linux/Flutter/generated_plugin_registrant.h
        modified:   FlutterProjects/my_first_flutter_app/linux/Flutter/generated_plugins.cmake
        modified:   FlutterProjects/my_first_flutter_app/macos/Flutter/GeneratedPluginRegistrant.swift
        modified:   FlutterProjects/my_first_flutter_app/windows/Flutter/generated_plugin_registrant.cc
        modified:   FlutterProjects/my_first_flutter_app/windows/Flutter/generated_plugin_registrant.h
        modified:   FlutterProjects/my_first_flutter_app/windows/Flutter/generated_plugins.cmake

no changes added to commit (use "git add" and/or "git commit -a")
```

satom@Tomilings MINGW64 ~/OneDrive - TAFE NSW/IT314AB_Verano (feature/conditional-ui)

```
$ git add .
warning: in the working copy of 'FlutterProjects/my_first_flutter_app/linux/Flutter/generated_plugin_registrant.cc', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'FlutterProjects/my_first_flutter_app/linux/Flutter/generated_plugin_registrant.h', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'FlutterProjects/my_first_flutter_app/linux/Flutter/generated_plugins.cmake', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'FlutterProjects/my_first_flutter_app/macos/Flutter/GeneratedPluginRegistrant.swift', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'FlutterProjects/my_first_flutter_app/windows/Flutter/generated_plugin_registrant.cc', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'FlutterProjects/my_first_flutter_app/windows/Flutter/generated_plugin_registrant.h', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'FlutterProjects/my_first_flutter_app/windows/Flutter/generated_plugins.cmake', LF will be replaced by CRLF the next time Git touches it
```

satom@Tomilings MINGW64 ~/OneDrive - TAFE NSW/IT314AB_Verano (feature/conditional-ui)

```
$ git commit -m "feat: submit Active or Inactive?"
[feature/conditional-ui 0da1ca] feat: submit Active or Inactive?
 1 file changed, 42 insertions(+), 23 deletions(-)
```

satom@Tomilings MINGW64 ~/OneDrive - TAFE NSW/IT314AB_Verano (feature/conditional-ui)

```
$ git push
Enumerating objects: 11, done.
Counting objects: 100% (11/11), done.
Delta compression using up to 20 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (6/6), 920 bytes | 920.00 KiB/s, done.
Total 6 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 3 local objects.
To https://github.com/KVerano/IT314AB_Verano.git
  57073e5..0da1ca  feature/conditional-ui -> feature/conditional-ui
```